



COURSE PROJECT REPORT

Subject: SWD392

– Ha Noi, January 2026 –

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Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

I. Overview

I.1 Project Information

- Project name: Gia Hoa Phát E-business System
- Project code: GHP
- Group name: SWD392-G2
- Software type: Web Application

I.2 Project Team

Full Name	Role	Email	Mobile
Nguyễn Danh Triệu	Leader	nguyendanhtrieu250305@gmail.com	0355425515
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Vu Thi Thuy	Member		
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I.3 Software Process(Sửa lại theo topic)

The Gia Hoa Phát E-business System project follows the Agile Scrum methodology. The project is divided into multiple sprints, allowing the development team to incrementally deliver business value while continuously collecting feedback from stakeholders.

Phase 1: Project Initiation and Requirement Analysis

Objectives

- Understand Gia Hoa Phát's current business operations.
- Identify system requirements and business rules.
- Define the project scope and priorities.

Activities

- Interview stakeholders and analyze existing workflows.
- Identify system actors:
 - Customer
 - Staff
 - Administrator
- Define functional and non-functional requirements.
- Create and prioritize the Product Backlog.

Deliverables

- Requirement Specification
 - Product Backlog
 - Initial Use Cases
-

Sprint 1: Product Catalog Management (2 Weeks)

Objectives

Develop the core product browsing functionality.

Features

- Product Category Management
- Product Information Management
- Product Search and Filtering
- Product Detail Display

Deliverables

- Product database structure
 - Product catalog interface
 - Product browsing functionality
-

Sprint 2: Customer Shopping Process (2 Weeks)

Objectives

Enable customers to purchase products online.

Features

- Shopping Cart Management
- Add/Remove Products from Cart
- Checkout Process
- Order Creation

Deliverables

- Shopping Cart Module
 - Checkout Workflow
 - Order Management Database
-

Sprint 3: Payment and Order Tracking (2 Weeks)

Objectives

Support payment processing and customer order monitoring.

Features

- Payment Integration
- Order Confirmation
- Order Status Tracking
- Customer Order History

Deliverables

- Payment Module
 - Order Tracking System
 - Customer Dashboard
-

Sprint 4: Inventory and Staff Management (2 Weeks)

Objectives

Support internal operational management.

Features

- Inventory Management
- Stock Quantity Updates
- Order Processing by Staff
- Warehouse Monitoring

Deliverables

- Inventory Module
 - Staff Management Functions
 - Stock Monitoring Dashboard
-

Sprint 5: Administration and Reporting (2 Weeks)

Objectives

Provide system administration and business monitoring tools.

Features

- User Management
- Product Management
- Sales Reporting
- Inventory Reporting

Deliverables

- Admin Dashboard
 - Reporting System
 - User Management Module
-

Sprint 6: Testing, Deployment, and Review (1 Week)

Objectives

Ensure system quality and prepare for production release.

Activities

- Unit Testing
- Integration Testing
- User Acceptance Testing (UAT)
- Bug Fixing
- Production Deployment

Deliverables

- Tested System
 - Deployment Package
 - Final Project Documentation
 - Sprint Review Report
-

Expected Outcome

After completing all sprints, Gia Hoa Phát will have a fully functional E-business system that supports online sales of baking ingredients and equipment, inventory management, order processing, and business administration.

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II. Requirement Specification

II.1 Problem description

Gia Hoa Phat is an established enterprise operating in the food and baking industry. Currently, the business relies heavily on a traditional, offline retail model through physical shops to sell baking ingredients (e.g., flour, butter, sugar) and baking equipment (e.g., ovens, baking molds). To adapt to the digital age and expand its market reach, the company is implementing a digital transformation project to shift from a traditional business model to a comprehensive e-business platform.

The "Gia Hoa Phat E-Business System" is a newly proposed software solution designed to centralize product cataloging, automate online purchasing, streamline inventory management across multiple branches, and facilitate seamless order fulfillment.

II.2 Major Features

FE-01: Browse, search, and filter a centralized online catalog containing both perishable baking ingredients (e.g., flour, butter) and heavy baking equipment (e.g., industrial ovens, mixers).

FE-02: Add items to a persistent shopping cart and securely check out using multiple payment methods, including Cash on Delivery (COD) and integrated third-party payment gateways (e.g., VNPay, MoMo).

FE-03: Manage and track real-time inventory levels dynamically across multiple physical store branches, including automated low-stock alerts for staff.

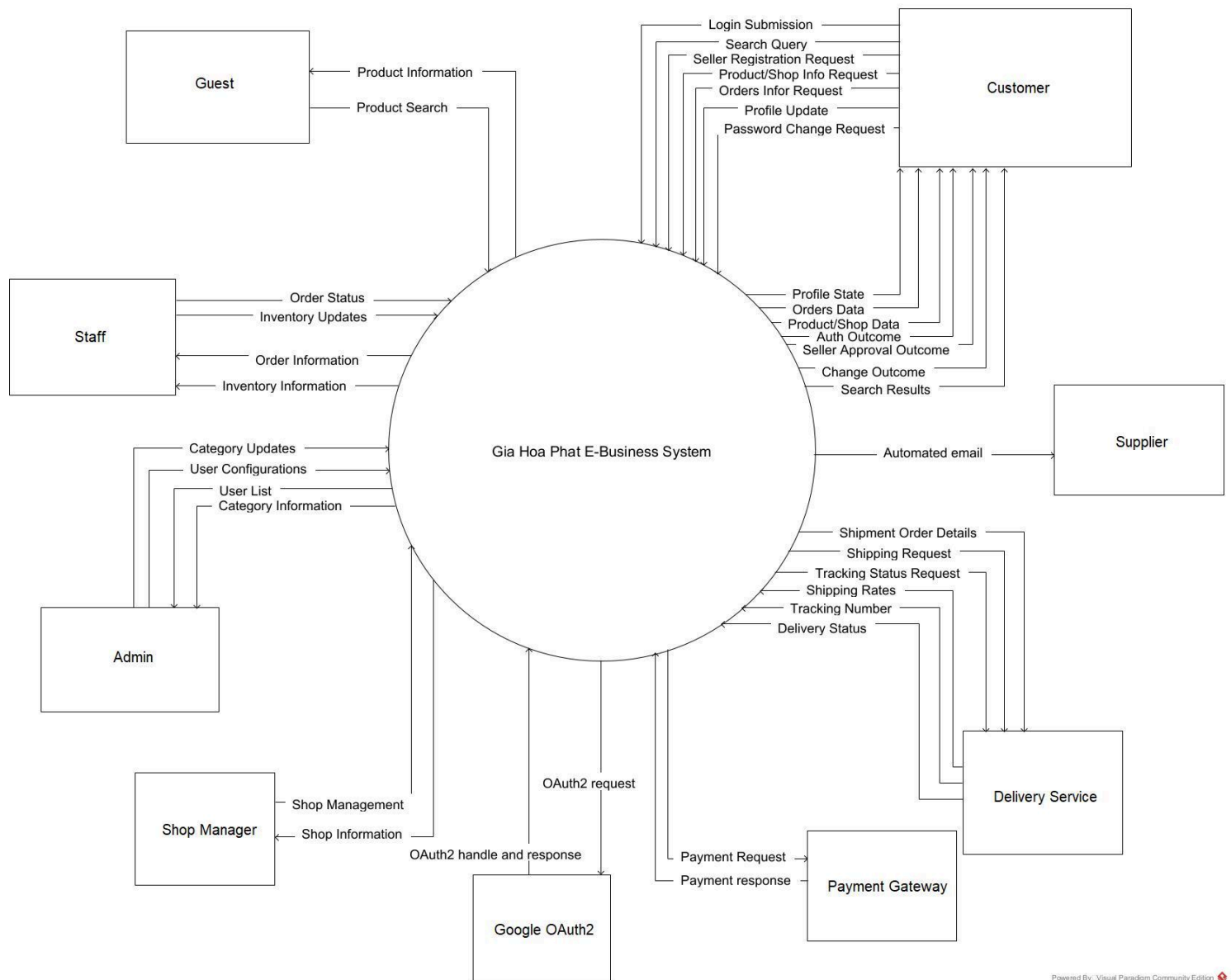
FE-04: Process and manage the full lifecycle of customer orders (Pending, Confirmed, Shipping, Completed, Cancelled), including a dedicated workflow for handling returns and refunds of technical equipment.

FE-05: Automate shipping logistics by integrating with third-party delivery services to generate waybills, calculate shipping fees based on item weight/size, and provide real-time tracking updates to customers.

FE-06: Provide a secure, role-based administrative dashboard for Staff and Admins to manage product listings, user accounts, and extract real-time revenue and sales reports.

FE-07: Enable customers to leave verified reviews and ratings for products they have purchased, while allowing staff to manage and respond to customer feedback.

II.3 Context Diagram



Powered By: Visual Paradigm Community Edition

II.4 Nonfunctional requirements

#	Feature	System Function	Description
1	Payment Integration	Payment Webhook Listener API	An endpoint that listens for asynchronous payment status updates (success/failure) from third-party payment gateways (e.g., VNPay, MoMo) and automatically updates the corresponding Order status in the database without manual staff intervention.
2	Inventory Management	Low Stock Alert Cron Job	A scheduled background batch job that runs nightly at 12:00 AM. It scans the Inventory table to identify baking ingredients or equipment that have fallen below their min_quantity threshold and triggers automated warning emails to the Store Manager.

3	Order Management	Auto-Cancel Unpaid Orders Job	A background service that runs every 30 minutes to check for orders with a "Pending Payment" status created more than 24 hours ago. It automatically changes their status to "Cancelled" and rolls back the reserved item quantities to the available stock.
	Shipping Management	Delivery Status Sync API	A system service that interacts with third-party delivery APIs (e.g., GHN, Viettel Post) to receive Webhook notifications or periodically fetch tracking updates, automatically updating the Shipment status (e.g., In-Transit, Delivered) in the local database.
	System Security & Performance	Expired Session Cleanup Batch	A scheduled task running once a week to clear expired user authentication tokens, OTP codes, and abandoned temporary shopping carts from the database to free up storage space and optimize query performance.
	Sales Reporting	Nightly Revenue Aggregation Service	A background batch job that runs at 2:00 AM daily to calculate and aggregate the total successful sales, shipping fees, and discounts from the previous day, storing the results in a separate reporting table for faster Admin dashboard loading.

II.5 Functional requirements

II.5.1 Actors

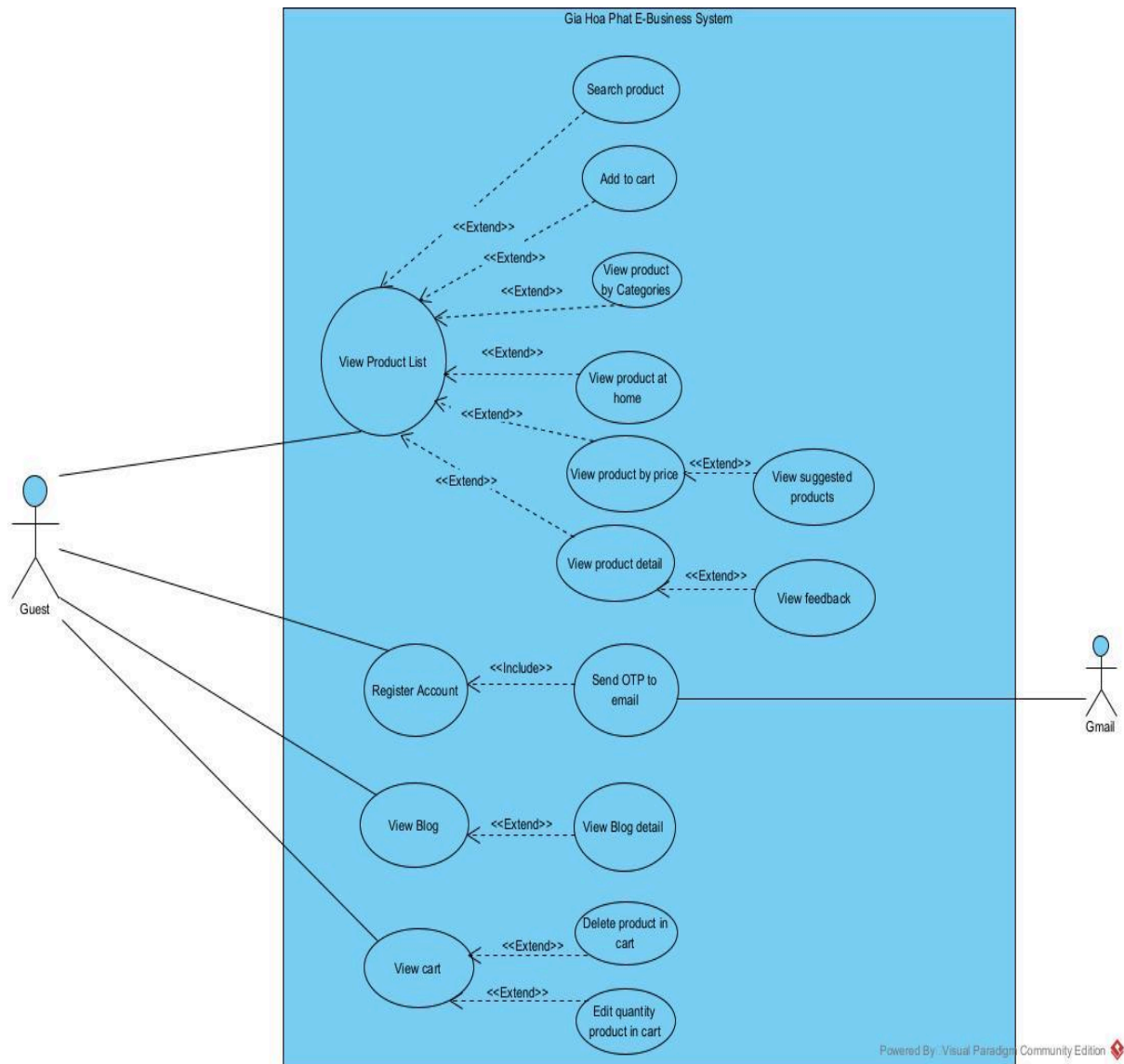
#	Actor	Description
1	Administrator	A human user with the highest global access privileges. Responsible for configuring system-wide settings, managing master product categories, overseeing all branch managers, and viewing cross-branch revenue reports.
2	Manager	A mid-level human user responsible for overseeing a specific physical branch. They manage local branch staff, view branch-specific sales and inventory reports, and handle escalated issues (e.g., complex refunds or stock discrepancies).
3	Staff	A human user responsible for daily operations within a specific branch. They interact with the system to process incoming orders, update the inventory levels of baking ingredients and equipment, and manage shipments.
4	Customer	The primary end-user of the e-business storefront. They use the system to browse the baking catalog, manage their shopping cart, place online orders, make payments, and track their delivery status.
5	Payment Gateway	An external system actor that provides financial services. It securely processes online customer transactions and notifies the e-business system of payment success or failure.

6	Delivery Service	An external system actor (e.g., GHN, Viettel Post) that interacts with the system to receive shipment requests and provide real-time tracking numbers and delivery status updates.
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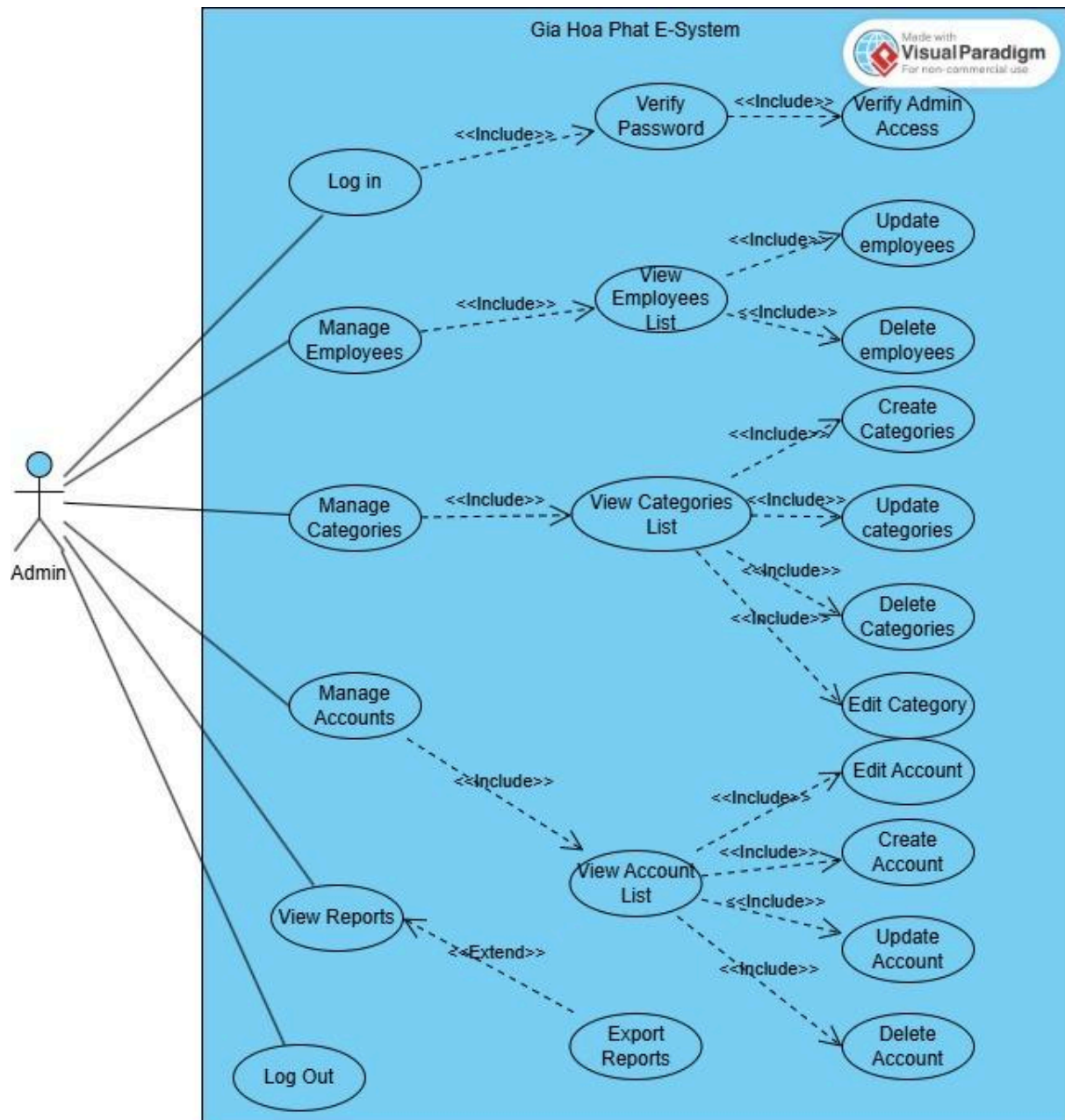
II.5.2 Use Cases

II.5.2 .1 Diagram(s)

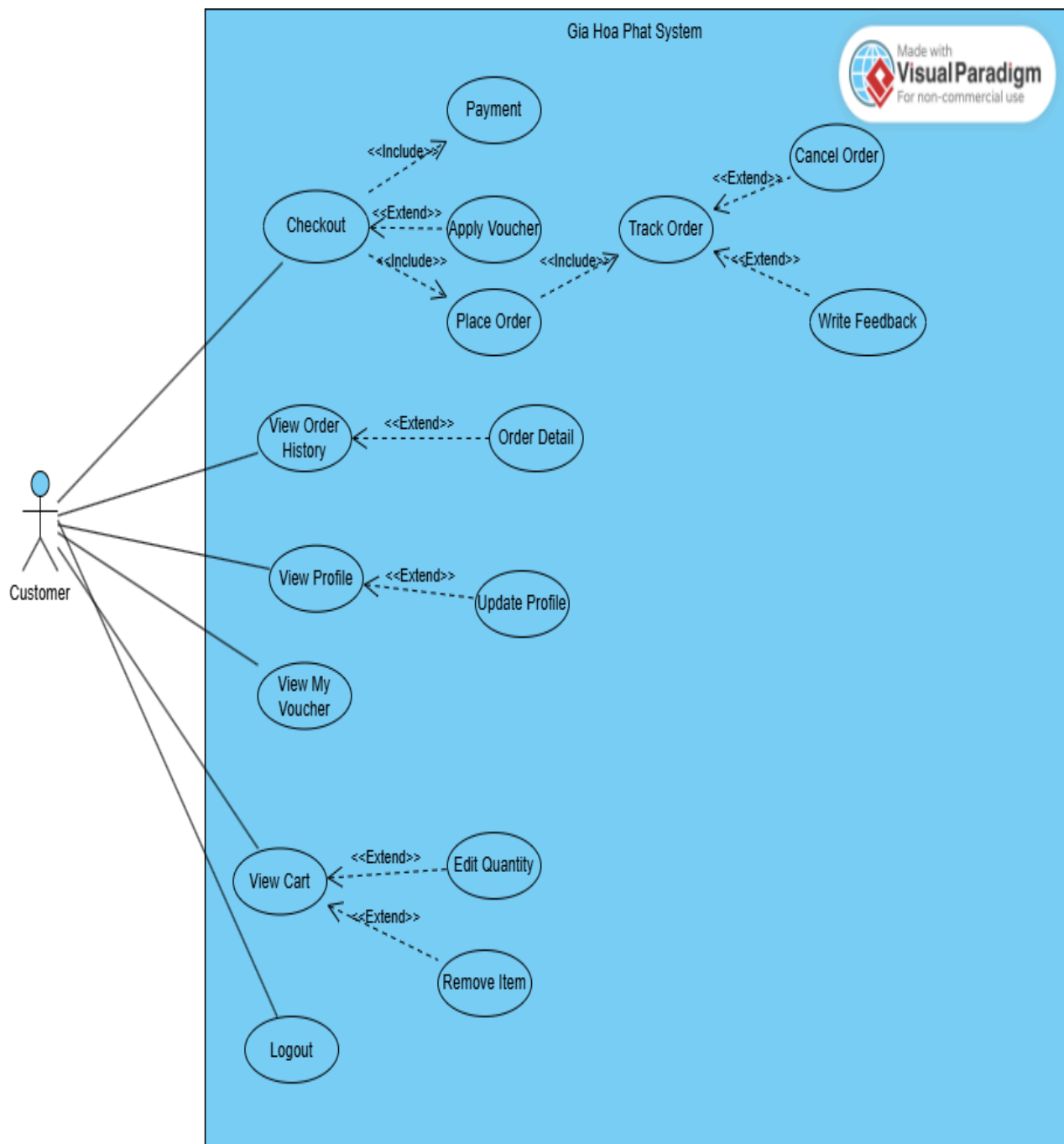
GUEST use case:



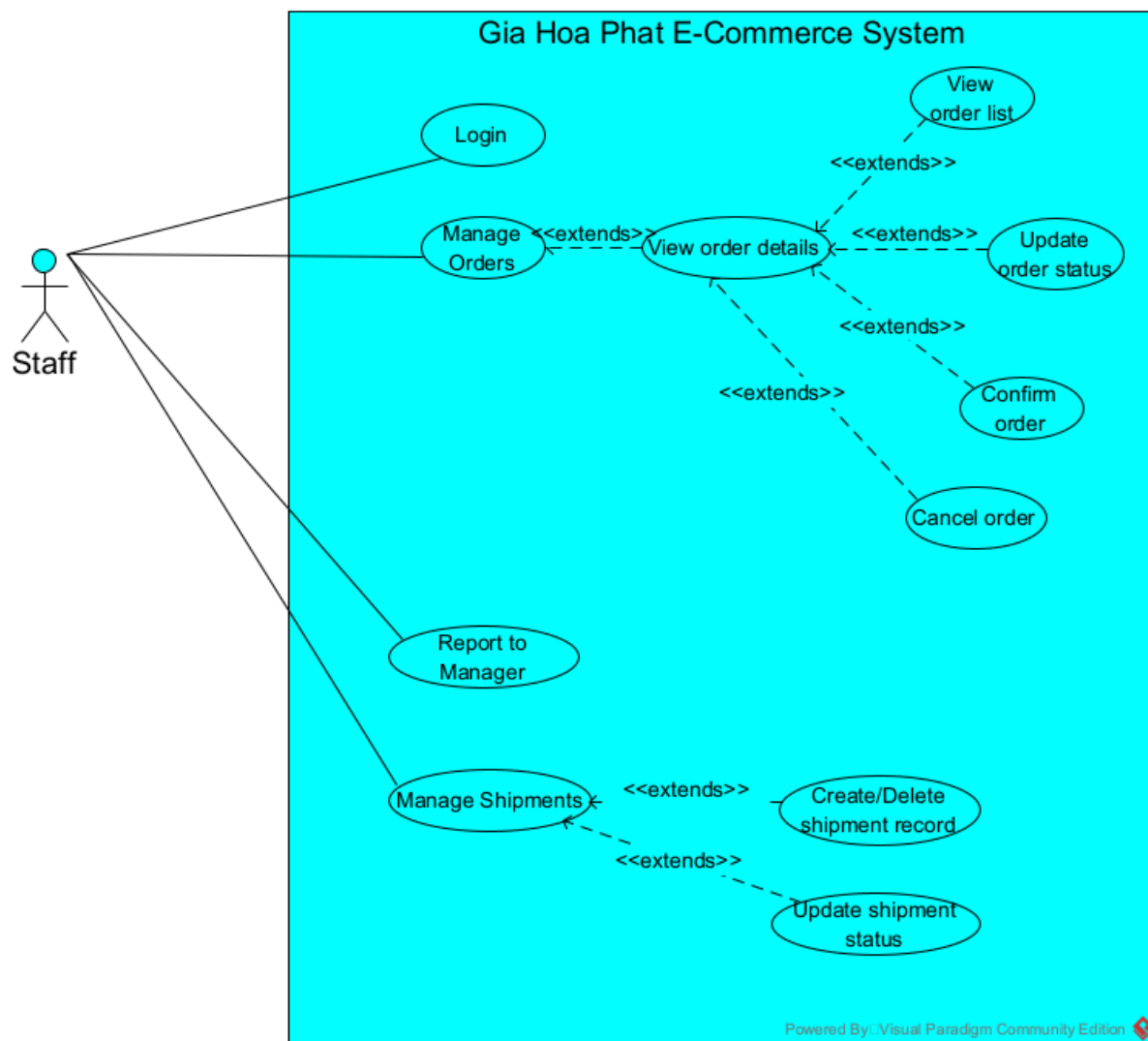
ADMIN Use case



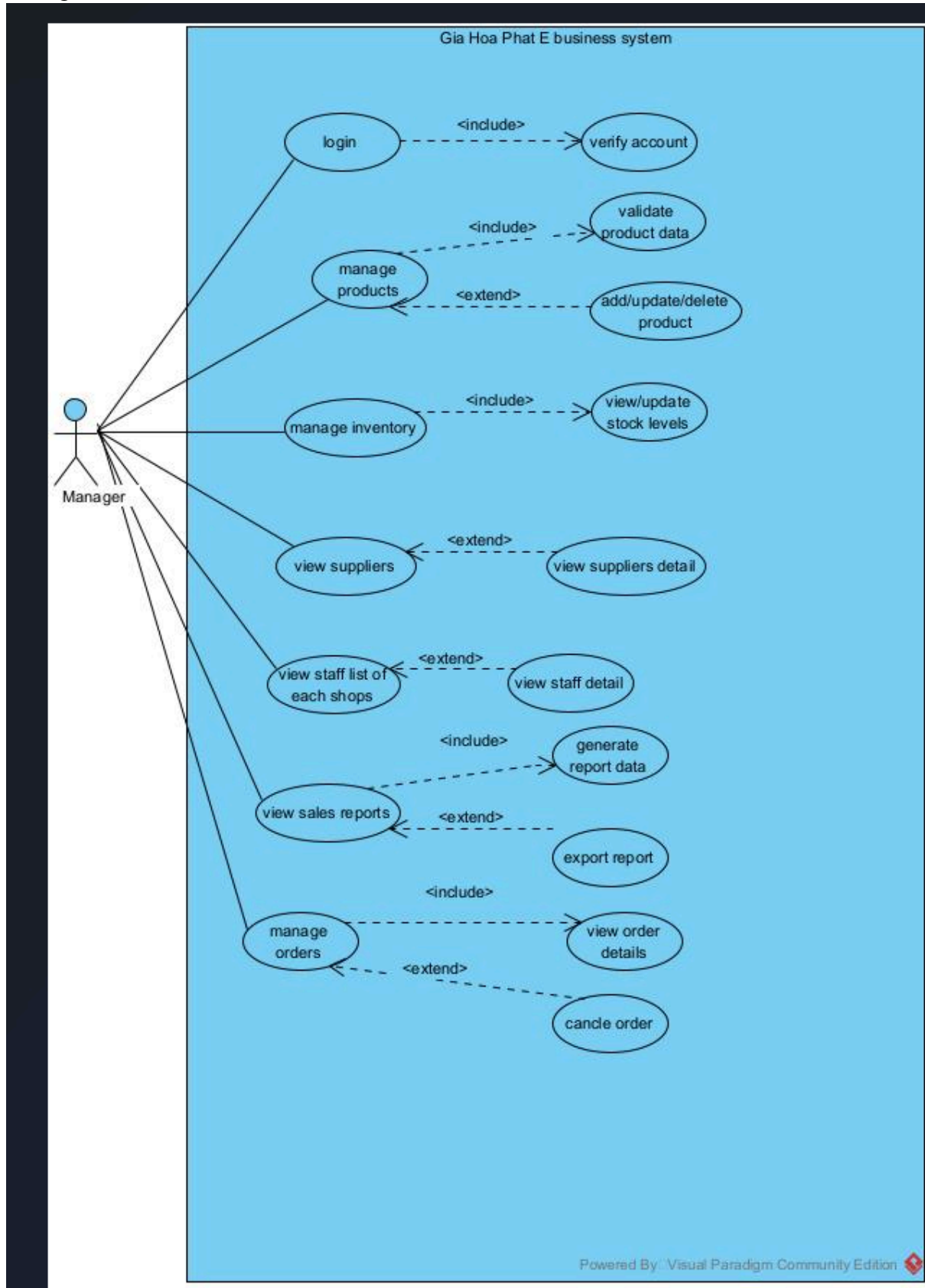
Use Case Customer



Staff Use Case Diagram



Manager Use case



II.5.2.2 Use case descriptions

This part describes the use cases, you can follow the table form as below/

ID	Use Case	Actors	Use Case Description
01	Withdraw Funds	ATM Customer	
02	Query Account	ATM Customer	
03	...		

1 <<UseCaseCode_UC Name>>

UC ID and Name:			
Created By:		Date Created:	
Primary Actor:		Secondary Actors:	
Trigger:			
Description:			
Preconditions:			
Postconditions:			
Normal Flow:			
Alternative Flows:			
Exceptions:			
Priority:	High (Medium, Low), Must Have (Should Have, Could Have),..		
Frequency of Use:			
Business Rules:			
Other Information:			
Assumptions:			

Functional Description Template

2.<UseCase01_Log In to System with Admin Credentials>

UC ID and Name:	UC-01 - Log in to System		
Created By:	DucHM	Date Created:	20/06/2026
Primary Actor:	Admin	Secondary Actors:	
Trigger:	Admin opens the system application or navigate to admin portal URL		
Description:	Allows the Admin to authenticate their credentials and gain access to the dedicated administrative dashboard.		
Preconditions:	The system is active, and the Admin possesses a pre-registered account in the database.		
Postconditions:	The Admin session is established, and the system unlocks the Admin Dashboard interface.		
Normal Flow:	1. The Admin enters their username and password into the login interface. 2. The Admin clicks the "Log in" button.		

	3. The system triggers the system-level validation process. 4. The system confirms the credentials and administrative rights. 5. The system initiates the secure session and grants access. 6. The system redirects the Admin to the main controller screen.
Alternative Flows:	AF01: Log in via Google Account (OAuth2): 1. On the login screen, the Admin clicks the "Sign in with Google" button. 2. The system redirects the Admin to the secure Google Account selection page. 3. The Admin authenticates via Google and grants profile access permissions. 4. The Google OAuth Service returns a secure access token containing the Admin's verified email address back to the system. 5. The system receives the token, checks the email against the internal database registry, and proceeds directly to step 4 of the Normal Flow (<<Include>> Verify Admin Access).
Exceptions:	Authentication / Privilege Failure at Step 3 & 4: 3.a/4.a. The system detects invalid credentials or a lack of administrative role 3.b/4.b. The system rejects the access request and displays a specific error notification 3.c/4.c. The system returns the workflow to the initial node for re-entry.
Priority:	High(Must have)
Frequency of Use:	High
Business Rules:	BR04: Accounts are automatically locked for 15 minutes after 5 consecutive failed login attempts to prevent brute-force attacks.
Other Information:	This Use Case implicitly encapsulates (<<include>>) the Verify Admin Privilege process executed by the backend system layer.
Assumptions:	

2.<UC02 - MANAGE EMPLOYEES>

UC ID and Name:	UC02 - Manage Employees		
Created By:	DucHM	Date Created:	20/06/2026

Primary Actor:	Admin	Secondary Actors:	
Trigger:	The Admin selects the employee management module on the dashboard.		
Description:	Provides administrative tools to create staff user accounts, configure profile fields, and assign system roles.		
Preconditions:	The Admin has successfully bypassed security checks and maintains an active session dashboard.		
Postconditions:	New user data is successfully updated and persistent in the storage tier.		
Normal Flow:	Sub-flow 1: View Employees List 1. The Admin hits the employee management entry point. 2. The system queries the user table database records. 3. The system maps and renders the target entities onto the View Employees List grid dashboard view. Sub-flow 2: Update Employees 1. From the generated list view, the Admin clicks the edit control modifier next to an employee entry. 2. The system initiates the sub-process view Update employees and loads a pre-populated form. 3. The Admin adjusts properties (e.g., changing role rankings or editing contact profiles) and saves. 4. The system validates constraints, updates the relational record, and updates the list viewport grid. Sub-flow 3: Delete Employees 1. From the list layout view, the Admin clicks the delete option button next to a target employee row. 2. The system executes the component routine Delete employees, rendering a verification prompt. 3. Upon Admin validation confirmation, the system terminates/purges the target user row or updates their structural active status flag in the repository storage.		
Alternative Flows:	None		
Exceptions:	Database Sync Timeout / Network Error:		

	1.a. The server encounters connection dropouts during update or deletion commits. 1.b. The system rolls back transactional commits to prevent corruption states and triggers an inline system alert banner to notify the Admin user.
Priority:	High, Must Have
Frequency of Use:	Low to Medium
Business Rules:	BR01: Employee login identifiers must be entirely unique across the entire user database registry. BR02: Every created account must be bound to at least one system permission profile prior to saving.
Other Information:	Form submission implicitly embeds the mandatory dependency rule to evaluate structural parameters via Validate Data.
Assumptions:	Connection channels to the secure authorization directory remain active during execution.

3.<UC03 - MANAGE CATEGORIES >

UC ID and Name:	UC03 - Manage Categories		
Created By:	DucHM	Date Created:	20/6/2026
Primary Actor:	Admin	Secondary Actors:	
Trigger:	The Admin clicks on the category management selection tab.		
Description:	Enables the Admin to register, structure, and modify inventory classification categories within the e-system.		
Preconditions:	Active operational state session with valid Admin Access verification privileges.		
Postconditions:	The categorization matrix is updated and safely written to storage.		
Normal Flow:	Sub-flow 1: View Category List & Search 1. The Admin selects the "Category Management" tab. 2. The system queries the database and renders the Category List View along with a global search input field. 3. The Admin can enter keywords to dynamically filter the category list. Sub-flow 2: Create New Category 1. The Admin clicks the "Add New Category" button from the list view. 2. The system opens the Create Category Modal Form.		

	3. The Admin enters the name and code, then clicks "Save". 4. The system executes the process. 5. The system commits the new record to the database and refreshes the list view. Sub-flow 3: Update Category 1. The Admin clicks the "Edit" icon next to a specific category row. 2. The system displays the Edit Category View with pre-filled existing values. 3. The Admin modifies the fields and clicks "Confirm Changes". 4. The system invokes <<Include>> Validate Data. If successful, updates the database. Sub-flow 4: Delete Category 1. The Admin clicks the "Delete" icon on a target category. 2. The system triggers a confirmation alert dialog ("Are you sure?"). 3. Upon Admin confirmation, the system updates the database status (or removes the record) and updates the UI grid view.
Alternative Flows:	None
Exceptions:	EX01: Form Validation Failure (Sub-flow 2 & 3): - The system detects duplicate category codes or blank fields during validation. - The system invokes the exception branch (<Extend>> Display error message) to prompt warnings on the active form.
Priority:	High, Must Have
Frequency of Use:	Low
Business Rules:	BR03: Category labels and shorthand identification tags cannot possess overlapping duplicates within database parameters.
Other Information:	Standard system execution depends strictly on the validation framework via its <<include> link.
Assumptions:	

4.<UC04 - VIEW REPORTS>

UC ID and Name:	UC04 - View Reports		
Created By:	DuchM	Date Created:	20/6/2026

Primary Actor:	Admin	Secondary Actors:	
Trigger:	The Admin selects the reporting/monitoring module from the main navigation menu.		
Description:	Provides the Admin with visibility into system metrics, summaries, and performance logs.		
Preconditions:	The Admin session is valid, authorized, and active.		
Postconditions:	Data metrics are rendered on the screen without modifying data records.		
Normal Flow:	1. The Admin clicks on the "View Reports" section. 2. The system queries the reporting data from the database layer. 3. The system structures the data records into readable charts and summaries. 4. The system displays the interactive data tables and metrics to the Admin.		
Alternative Flows:	Optional Export Flow (Triggered by Admin choice): 4.a. While viewing the metrics, the Admin decides to trigger local file saving. 4.b. The system activates the optional feature extension (<<Extend>> Export Reports). 4.c. The system compiles the active report dataset into a downloadable format (e.g., CSV, Excel, PDF).		
Exceptions:	Database Connection Timeout: 2.a. The database fails to return data logs within the designated network timeout window. 2.b. The system displays a connection warning and falls back to a blank dashboard state.		
Priority:	Medium, Should Have		
Frequency of Use:	Regularly		
Business Rules:	BR05: Loaded reporting metrics must accurately match current database records.		
Other Information:	The Export Reports use case is completely dependent on this base view, executing conditionally via an <<Extend>> line.		
Assumptions:			

5. UC Register User Account

UC ID and Name:	UC04 - Register User Account		
Created By:	LinhNTN	Date Created:	21/6/2026
Primary Actor:	Guest	Secondary Actors:	Google Gateway
Trigger:	A guest decides to join the system and make an account.		
Description:	A guest creates a new account using either their email and password or their Google account. This allows them to log in and access the system's features.		
Preconditions:	PRE-1: The email address is not yet used in the system. PRE-2: The website is able to connect to Google's login service.		
Postconditions:	POST-1: A new user profile is saved in the database. POST-2: The user is logged in automatically right after signing up.		
Normal Flow:	1. The guest clicks the "Sign Up" button. 2. The website shows a form asking for an email and password. 3. The guest types in their details and submits the form. 4. The website checks if the email is new and the password is safe. 5. An email with a secret code (OTP) is sent to the guest. 6. The guest types this code into the website. 7. The system saves the details and creates the account. 8. The guest becomes a "Customer", logs in automatically, and is sent to the homepage.		
Alternative Flows:	1. The guest clicks "Sign up with Google". 2. The website sends the guest to Google's login screen. 3. The guest logs into their Google account. 4. The website gets the guest's basic info (like name and email) from Google.		

	5. The new account is created and activated right away. 6. The guest is logged into the system automatically.
Exceptions:	E1: Database is down -> Cannot finish signing up. E2: Email system is down -> Cannot send the OTP code, so signing up stops. E3: Email is already used. E4: Wrong OTP code typed. E5: Google login fails.
Priority:	High
Frequency of Use:	High
Business Rules:	None
Other Information:	Users can only ask to resend the OTP code a few times to stop spamming.
Assumptions:	- The user can open and read the email they used to sign up. - Google and the email systems are working correctly.

4.<UC06 - Manage Accounts>

UC ID and Name:	UC06 - Manage Accounts		
Created By:	DucHM	Date Created:	20/6/2026
Primary Actor:	Admin	Secondary Actors:	
Trigger:	The Admin clicks the "Manage Accounts" system configurations menu link on the control dashboard		
Description:	Comprehensive subsystem permitting the Admin to access secure account entities, provision new system credentials, edit access parameters, and revoke login permissions.		
Preconditions:	Secure portal login state is verified, and session tokens remain fully valid.		
Postconditions:	Security parameters, login states, and account access profiles are updated within persistent storage.		
Normal Flow:	Sub-flow 1: View Account List 1. The Admin initiates the account infrastructure module controller.		

	2. The system interfaces the secure identity layer and builds the View Account List overview viewport grid. Sub-flow 2: Create Account 1. The Admin clicks the "Create New Account" action trigger. 2. The system opens the Create Account view portal container form fields. 3. The Admin provides critical security indicators (Username, Security Group, Password) and hits submit. 4. The system checks syntax formats, hashes structural keys, saves the node item, and returns a success pop-up notification. Sub-flow 3: Update Account 1. The Admin clicks the credentials edit option link for a selected entry record. 2. The system triggers Update Account, bringing forward properties (Account Status flags, Reset Token triggers). 3. The Admin applies updates and commits changes to the host engine interface view. Sub-flow 4: Delete Account 1. The Admin triggers the deletion trashcan icon next to a specific active target line field item. 2. The system engages the Delete Account security subsystem modal framework interface wrapper. 3. The Admin confirms the operation, prompting the storage tier to invalidate access keys.
Alternative Flows:	None
Exceptions:	Credential Complexity Requirement Failure (Sub-flow 2 & 3): 3.a. The entered system password falls short of structural cryptographic enterprise security parameters. 3.b. The system blocks database submission requests and prompts a validation error string to the active form screen layout view.
Priority:	High, Must Have

Frequency of Use:	Medium
Business Rules:	BR07: Password entries must meet security policies prior to structural execution phases. BR08: System identity indices must remain strictly unique to prevent session validation clashes.
Other Information:	Operations map cleanly across sequential actions tied straight into the core security database tables.
Assumptions:	Encryption algorithm modules (such as BCrypt/Argon2) are properly loaded at the application layer server.

II.5.2.2.2 Manager Use case descriptions

1.UC-M01:Manage Products

UC ID and Name:	UC-M01 - Manage Products		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects "Product Management" from menu		
Description:	Managers can add new products, edit existing products, or delete products from the system. Each product includes name, description, price, category, stock quantity, and images		
Preconditions:	Manager is logged in and authenticated		
Postconditions:	Product information is updated in the database		
Normal Flow:	1. Manager selects "Add/Edit/Delete Product" 2. System displays product list 3. Manager selects a product or adds new 4. System validates product information 5. System saves changes to Database 6. System confirms success		
Alternative Flows:	3a. If product already exists, manager updates information instead		
Exceptions:	4a. Invalid data (negative price, empty name) → System shows error message 5a. Database connection fails → System displays "Cannot connect to database"		
Priority:	High, Must Have		
Frequency of Use:	Daily (multiple times)		
Business Rules:	Product code must be unique; Price must be > 0; Stock quantity cannot be negative		
Other Information:	Products are organized by categories (ingredients, tools, equipment)		
Assumptions:	Manager has full CRUD permissions on products		

2.UC-M02:Manage Inventory

UC ID and Name:	UC-M02 - Manage Inventory		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects "Manage Inventory"		
Description:	Manager views current stock levels, adjusts inventory, and sets reorder thresholds for baking ingredients and equipment.		
Preconditions:	Manager is logged in and authenticated		
Postconditions:	Inventory records are updated		
Normal Flow:	1. Manager selects Manage Inventory 2. System <> Checks Stock Levels 3. System displays current inventory 4. Manager adjusts stock quantities 5. System validates input 6. System update and confirms		
Alternative Flows:	<<extend>> UC-M01 Manage Products : Manager can update stock directly from product management page		
Exceptions:	4a. Stock quantity adjustment exceeds reasonable limits → System shows warning		
Priority:	High, Must Have		
Frequency of Use:	Daily		
Business Rules:	Low stock alert when quantity < reorder threshold; Negative stock not allowed		
Other Information:	Inventory is updated automatically when orders are placed		
Assumptions:	Physical count is done periodically to match system records		

3.UC-M03:Manage Orders

UC ID and Name:	UC-M03 - Manage Orders		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects "Manage Orders"		
Description:	Manager views all customer orders, checks order details, and updates order status (processing, shipping, completed, cancelled).		
Preconditions:	Manager is logged in; Orders exist in the system		
Postconditions:	Order status is updated		
Normal Flow:	1. Manager selects Manage Orders 2. System displays order list 3. Manager selects an order 4. System displays order details 5. Manager updates order status 6. System saves to Database 7. System confirms		
Alternative Flows:	<> UC-M04 Cancel Order: Manager cancels the order if needed		

UC ID and Name:	UC-M03 - Manage Orders		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Exceptions:	5a. Order already completed → Cannot change status 6a. Database error → Rollback changes		
Priority:	High, Must Have		
Frequency of Use:	Daily (multiple times)		
Business Rules:	Once order status is "Completed", it cannot be changed; Cancelled orders trigger inventory restock		
Other Information:	Order status flow: Pending → Confirmed → Packing → Shipping → Completed		
Assumptions:	Orders are created by customers via the online store		

4.UC-M04:Cancel Order

UC ID and Name:	UC-M04 - Cancel Order		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager clicks "Cancel Order"<<extend>> UC-M03 Manage Orders		
Description:	Manager cancels a customer order, refunds payment (if applicable), and restocks inventory.		
Preconditions:	Order exists and is not yet completed		
Postconditions:	Order status = "Cancelled"; Inventory is restored; Refund is processed		
Normal Flow:	1. Manager selects Cancel Order from Manage Orders 2. System asks for cancellation reason 3. Manager enters reason 4. System processes cancellation 5. System updates Order status to "Cancelled" 6. System restores inventory quantities 7. System processes refund (if paid) 8. System confirms success		
Alternative Flows:	6a. Order was COD (Cash on Delivery) → Skip refund step		
Exceptions:	2a. Manager does not provide reason → System warns but allows		
Priority:	Medium, Should Have		
Frequency of Use:	Occasionally (a few times per week)		
Business Rules:	Only orders with status "Pending" or "Confirmed" can be cancelled; Refund goes to original payment method		
Other Information:	Cancellation reason is recorded for reporting		
Assumptions:	Payment gateway supports refund API		

5. UC-M05:View staff list of each shop

UC ID and Name:	UC-M05 - View Staff List		
Created By:	tienpg	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects " View Staff" from menu		
Description:	Manager views the list of staff members for each shop, including their name, role, and contact information.		
Preconditions:	Manager is logged in and authenticated		
Postconditions:	Staff list is displayed		
Normal Flow:	1. Manager selects Staff 2. System displays staff list 3. Manager views staff details 4. System displays staff details.		
Alternative Flows:	Manager selects staff member → System displays that staff member's details.		
Exceptions:	No staff records found → System displays "No staff available."		
Priority:	Medium, Must Have		
Frequency of Use:	Weekly (when hiring new staff)		
Business Rules:	Each staff must have unique username and email; Roles determine system permissions		
Other Information:	Staff roles: Shipper, Customer Support		
Assumptions:	Staff list is maintained by system		

6.UC - M06: View Sales Reports

UC ID and Name:	UC-M06 - View Sales Reports		
Created By:	tienpg	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects "Sales Reports" from menu		
Description:	Manager views sales data including daily revenue, best-selling products, revenue by category, and order statistics. Charts and tables are displayed for analysis.		
Preconditions:	Manager is logged in; Sales data exists		
Postconditions:	Report is displayed on screen		
Normal Flow:	1. Manager selects View Sales Reports 2. System generates report data. 3. System displays report with charts and tables 4. Manager can filter by date range, category, product		

UC ID and Name:	UC-M06 - View Sales Reports		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
	5. System refreshes report		
Alternative Flows:	<<extend>> UC-M07 Export Report : Manager exports report to PDF/Excel		
Exceptions:	2a. No data for selected period → System shows "No data available"		
Priority:	High, Must Have		
Frequency of Use:	Daily (multiple times)		
Business Rules:	Reports are based on completed orders only; Data is aggregated by day/week/month		
Other Information:	Report types: Revenue report, Product performance, Category analysis		
Assumptions:	All sales transactions are recorded in the database		

7.UC-M07:Export Report

UC ID and Name:	UC-M07 - Export Report		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager clicks "Export" from View Sales Reports		
Description:	Manager exports the current sales report as PDF or Excel file for offline analysis or sharing with stakeholders.		
Preconditions:	Report data is displayed on screen		
Postconditions:	Report file is downloaded to local machine		
Normal Flow:	1. Manager selects export format (PDF/Excel) 2. System generates file 3. System prompts file download 4. Manager saves file locally		
Alternative Flows:	1a. Manager selects email option → System emails report directly		
Exceptions:	2a. File generation fails → System error message 3a. Download cancelled → No action taken		
Priority:	Low, Could Have		
Frequency of Use:	Weekly		
Business Rules:	Exported reports include report title, date range, and company logo		
Other Information:	PDF includes charts; Excel includes raw data tables		
Assumptions:	Local machine has PDF reader or Excel installed		

8.UC-M08:View Suppliers

UC ID and Name:	UC-M08 - View Suppliers		
Created By:	tienpq	Date Created:	22/6/2026
Primary Actor:	Manager	Secondary Actors:	None
Trigger:	Manager selects "View Supplier"		
Description:	Manager views supplier information.		
Preconditions:	Manager is logged in and authenticated		
Postconditions:	Supplier information is displayed		
Normal Flow:	1. Manager selects View Suppliers. 2. The system displays the supplier list. 3. Manager views supplier details		
Alternative Flows:	<<extend>> UC-M02 – Manage Inventory: Manager views suppliers while restocking inventory.		
Exceptions:	No supplier data → System displays an error message.		
Priority:	Medium, Should Have		
Frequency of Use:	Monthly (when sourcing new ingredients)		
Business Rules:	Supplier phone and email must be provided; One supplier can provide multiple products		
Other Information:	Suppliers provide baking ingredients, packaging materials, and equipment		
Assumptions:	Supplier list is maintained manually by Manager		

UC ID and Name:	UC-CUS-01: Checkout		
Created By:	ThangNT	Date Created:	20/06/2026
Primary Actor:	Customer	Secondary Actors:	
Trigger:	The customer clicks the checkout button from the shopping cart.		
Description:	This use case allows a customer to purchase products in the shopping cart. The customer reviews the order information, optionally applies a voucher, selects a payment method, and confirms the purchase. After a successful checkout, the system creates a new order for the customer.		
Preconditions	Before Checkout can begin, • the customer must log in, • the shopping cart must contain at least one product, • products must still be available,		

UC ID and Name:	UC-CUS-01: Checkout		
Created By:	ThangNT	Date Created:	20/06/2026
Primary Actor:	Customer	Secondary Actors:	
	and the customer must provide a shipping address.		
Postconditions:	Success: - Order is successfully created. - Inventory quantity is updated. - Shopping cart is emptied. - Customer can view the order in Order History and track its status. Failure: - No order is created. - Shopping cart contents remain unchanged		
Normal Flow:	- Customer opens the View Cart page. - Customer reviews all selected products. - Customer clicks Checkout. - System displays the Checkout page, including shipping information, order summary, and available payment methods. - Customer selects a payment method. - System performs the Payment process - After successful payment, customer confirms the purchase. - System performs Place Order. - System generates a new Order ID. - System updates product inventory. - System clears all purchased items from the shopping cart. - System displays a successful checkout message. - Customer can later access View Order History to monitor the order.		
Alternative Flows:	AF1 – Apply Voucher At Step 5, customer chooses Apply Voucher. The system displays all available vouchers. Customer selects one voucher. System validates the voucher and recalculates the total amount before continuing to payment. AF2 – Customer Cancels Checkout At Step 5, customer decides not to continue. System returns to the shopping cart. AF3 – Customer Changes Payment Method Before payment is completed, customer selects another payment method. System updates the payment information.		
Exceptions:	E1 – Payment Failed The payment gateway rejects the payment. System displays an error message. Customer may retry or choose another payment method.		

UC ID and Name:	UC-CUS-01: Checkout		
Created By:	ThangNT	Date Created:	20/06/2026
Primary Actor:	Customer	Secondary Actors:	
	E2 – Product Out of Stock Before creating the order, one or more products become unavailable. System informs the customer and requests an update to the shopping cart. E3 –System Error The system cannot create the order because of an internal error. An error message is displayed, and no order is created.		
Priority:	High (Medium, Low), Must Have (Should Have, Could Have),..		
Frequency of Use:	Very High. Executed whenever a customer purchases products.		
Business Rules:	BR-01: Customer must log in before performing Checkout. BR-02: Shopping cart must contain at least one product. BR-03: Payment must be completed successfully before placing an order BR-04: Voucher must be valid, active, and not expired. BR-05: A voucher can only be applied once per order. BR-06: Ordered quantity must not exceed the available stock. BR-07: System generates a unique Order ID for every successful order.		
Other Information:	The system supports multiple payment methods (e.g., Cash on Delivery and Online Payment). After a successful checkout, the customer can track, cancel , and review the order from View Order History .		
Assumptions:	1. Customer has a stable Internet connection. 2. Payment Gateway is operational. 3. Inventory data is synchronized with the database. 4. Shipping service is available for the selected address. 5. Customer provides accurate shipping information.		

II.5.2.2.3 Manage Orders

UC ID and Name:	UC-01 Manage Orders		
Created By:	TrieuND	Date Created:	22/6/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
Trigger:	The Staff clicks on the "Order Management" tab from the admin dashboard.		
Description:	This use case allows the Staff to view the list of customer orders, review order details, confirm new orders, update delivery statuses, or cancel an order if necessary.		

UC ID and Name:	UC-01 Manage Orders		
Created By:	TrieuND	Date Created:	22/6/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
Preconditions:	PRE-1: The Staff's identity has been successfully authenticated (Logged in). PRE-2: The Staff account has been granted the 'Order Management' permission.		
Postconditions:	POST-1: The state of the selected order is successfully updated in the database (e.g., Confirmed, Shipping, Cancelled). POST-2: Inventory quantity is adjusted if an order is cancelled.		
Normal Flow:	- Staff selects the "Order Management" menu. - System retrieves and displays the current list of orders (View order list extension). - Staff selects a specific order with "Pending" status to view its details. - The system displays the details of the selected order (items, shipping info, total amount). - Staff verifies the information and clicks the "Confirm Order" button. - System updates the order status to "Confirmed" in the database. - The system displays a success message and reloads the updated order list.		
Alternative Flows:	1.1 Update Order Status - In step 5 of the normal flow, the Staff clicks the "Update Status" button. - System displays a dropdown with available statuses (e.g., Shipping, Delivered). - Staff selects the appropriate status and clicks "Save". - The system updates the status in the database and shows a success message.		

UC ID and Name:	UC-01 Manage Orders		
Created By:	TrieuND	Date Created:	22/6/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
	1.2 Cancel Order 1. In step 5 of the normal flow, the Staff clicks the "Cancel Order" button. 2. The system prompts the Staff to input a cancellation reason. 3. Staff inputs the reason and confirms. 4. System updates the order status to "Cancelled" and rolls back the reserved inventory quantity (Update inventory quantity).		
Exceptions:	1.0.E1 Database Connection Timeout At step 2, if the system cannot retrieve data from the database, the system displays an error message: "Unable to load orders. Please try again later," and logs the error. 1.0.E2 Concurrent Modification Error At step 6 (Normal Flow) or step 4 (Alt Flows), if the system detects that the order status has already been changed by another Staff/Admin in the meantime, the system aborts the update and displays: "This order has been modified by another user. Please refresh the page."		
Priority:	High, Must Have		
Frequency of Use:	High (Expected to be used dozens of times per day by the Staff).		
Business Rules:	BR-01: Only orders with a "Pending" status can be Confirmed or Cancelled. BR-02: Once an order status is marked as "Delivered", it cannot be reverted to "Shipping" or "Pending".		
Other Information:	If an order is cancelled, the system should trigger an automated email notification to the Customer explaining the cancellation reason.		
Assumptions:	It is assumed that the Staff understands the internal business rules regarding when an order should be confirmed or cancelled.		

II.5.2.2.4 Manage Shipments

UC ID and Name:	UC-03 Manage Shipments		
Created By:	TrieuND	Date Created:	22/06/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
Trigger:	The Staff clicks on the "Shipment Management" tab from the dashboard.		
Description:	This use case allows the Staff to handle logistics by creating new shipment records for confirmed orders, updating their delivery statuses, or deleting shipment records if errors occur.		
Preconditions:	PRE-1: The Staff is logged in with shipment management permissions. PRE-2: A customer order must exist and be in the "Confirmed" or "Processing" state.		
Postconditions:	POST-1: A shipment record is successfully created, updated, or deleted in the database. POST-2: The associated order's status is automatically synced if the shipment is completed.		
Normal Flow:	1. Staff selects the "Manage Shipments" menu. 2. The system displays a list of pending orders waiting to be shipped and existing shipment records. 3. Staff selects a pending order and clicks "Create Shipment". 4. The system generates a form auto-filled with the customer's delivery details. 5. Staff selects the carrier, inputs the tracking code provided by the carrier, and clicks "Save". 6. The system validates the input and saves the new shipment record to the database.		

UC ID and Name:	UC-03 Manage Shipments		
Created By:	TrieuND	Date Created:	22/06/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
Trigger:	The Staff clicks on the "Shipment Management" tab from the dashboard.		
	7. System updates the associated order status to "Shipping". 8. The system displays a success message.		
Alternative Flows:	3.1 Update Shipment Status 1. In step 3 of the normal flow, the Staff selects an existing shipment record and clicks "Update Status". 2. System displays a dropdown with statuses (e.g., In Transit, Delivered, Returned). 3. Staff selects the new status and clicks "Confirm". 4. The system updates the shipment status in the database and shows a success message. 3.2 Delete Shipment Record 1. In step 3 of the normal flow, the Staff selects an existing shipment record and clicks "Delete". 2. The system prompts a confirmation dialog: "Are you sure you want to delete this shipment?" 3. Staff clicks "Yes". 4. System removes the shipment record from the database and reverts the associated order status back to "Confirmed".		
Exceptions:	3.0.E1 Missing Tracking Code At step 6 (Normal Flow), if the tracking code is left blank, the system prevents saving and displays: "Tracking code is required to create a shipment." 3.2.E1 Cannot Delete Active Shipment		

UC ID and Name:	UC-03 Manage Shipments		
Created By:	TrieuND	Date Created:	22/06/2026
Primary Actor:	Staff	Secondary Actors:	System, Database
Trigger:	The Staff clicks on the "Shipment Management" tab from the dashboard.		
	At step 4 (Alt Flow 3.2), if the shipment status is already "In Transit" or "Delivered", the system blocks the deletion and displays: "Cannot delete a shipment that is already in progress or completed."		
Priority:	Medium, Should Have		
Frequency of Use:	High (Multiple times a day as orders are packed and dispatched).		
Business Rules:	BR-01: Every shipment record must be linked to a valid Order ID. BR-02: If a shipment status is updated to "Delivered", the parent Order must automatically be marked as "Completed".		
Other Information:	Creating a shipment or updating its status to "Delivered" should trigger an automated email to the customer with their tracking information.		
Assumptions:	The external delivery service (e.g., Viettel Post, GHN) provides accurate tracking codes to the Staff during the packing process.		

Functional Description Contents

Use Case ID and Name

Give each use case a unique integer sequence number identifier. State a concise name for the use case that indicates the value the use case would provide to some user. Begin with an action verb, followed by an object.

Author and Date Created

Enter the name of the person who initially wrote this use case and the date it was written.

Primary and Secondary Actors

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the primary actor that will be initiating this use case and any other secondary actors who will participate in completing execution of the use case.

Trigger

Identify the business event, system event, or user action that initiates the use case. This trigger alerts the system that it should begin testing the preconditions for the use case so it can judge whether to proceed with execution.

Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. The system must be able to test each precondition. Number each precondition. Example: PRE-1: User's identity has been authenticated.

Postconditions

Describe the state of the system at the successful conclusion of the use case execution. Label each postcondition in the form POST-X, where X is a sequence number. Example: POST-1: Price of item in the database has been updated with the new value.

Normal Flow

Provide a description of the user actions and corresponding system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. Show a numbered list of actions performed by the actor, alternating with responses provided by the system. The normal flow is numbered "X.0", where "X" is the Use Case ID.

Alternative Flows

Document other successful usage scenarios that can take place within this use case. State the alternative flow, and describe any differences in the sequence of steps that take place. Number each alternative flow in the form "X.Y", where "X" is the Use Case ID and Y is a sequence number for the alternative flow. For example, "5.3" would indicate the third alternative flow for use case number 5. Indicate where each alternative flow would branch off from the normal flow, and if pertinent, where it would rejoin the normal flow.

Exceptions

Describe any anticipated error conditions that could occur during execution of the use case and how the system is to respond to those conditions. Number each alternative flow in the form "X.Y.EZ", where "X" is the Use Case ID, Y indicates the normal (0) or alternative (>0) flow during which this exception could take place, "E" indicates an exception, and "Z" is a sequence number for the exceptions. For example "5.0.E2" would indicate the second exception for the normal flow for use case number 5. Indicate where in the normal (or an alternative) flow each exception could occur.

Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. Use the same priority scheme as that used for the functional requirements.

Frequency of Use

Estimate the number of times this use case will be performed per some appropriate unit of time. This gives an early indicator of throughput, concurrent usage loads, and transaction capacity.

Business Rules

List any business rules that influence this use case. Don't include the business rule text here, just its identifier so the reader can find it in another repository when needed.

Other Information

Identify any additional requirements, such as quality attributes, for the use case that may need to be addressed during design or implementation. Also list any associated functional requirements that aren't a direct part of the use case flows but which a developer needs to know about. Describe what should happen if the use case execution fails for some

unanticipated or systemic reason (e.g., loss of network connectivity, timeout). If the use case results in a durable state change in a database or the outside world, state whether the change is rolled back, completed correctly, partially completed with a known state, or left in an undetermined state as a result of the exception.

Assumptions

List any assumptions that were made regarding this use case or how it might execute.

b. Business Rules

Provide the business rules those are applied only to the use case

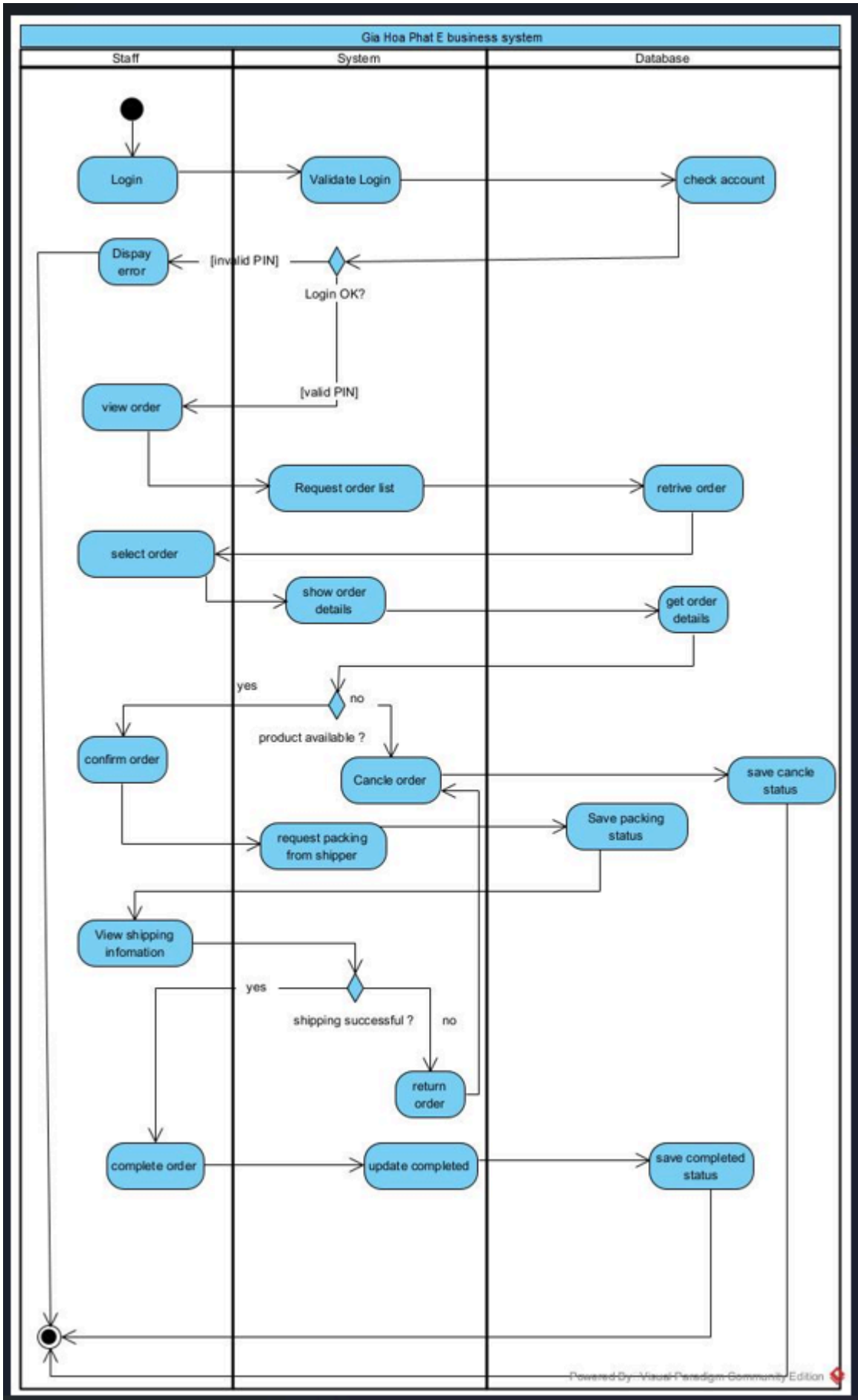
ID	Business Rule	Business Rule Description
FR1	Password Encoding	User's password must be encoded with MD5 hashing

II.5.3 Activity diagram

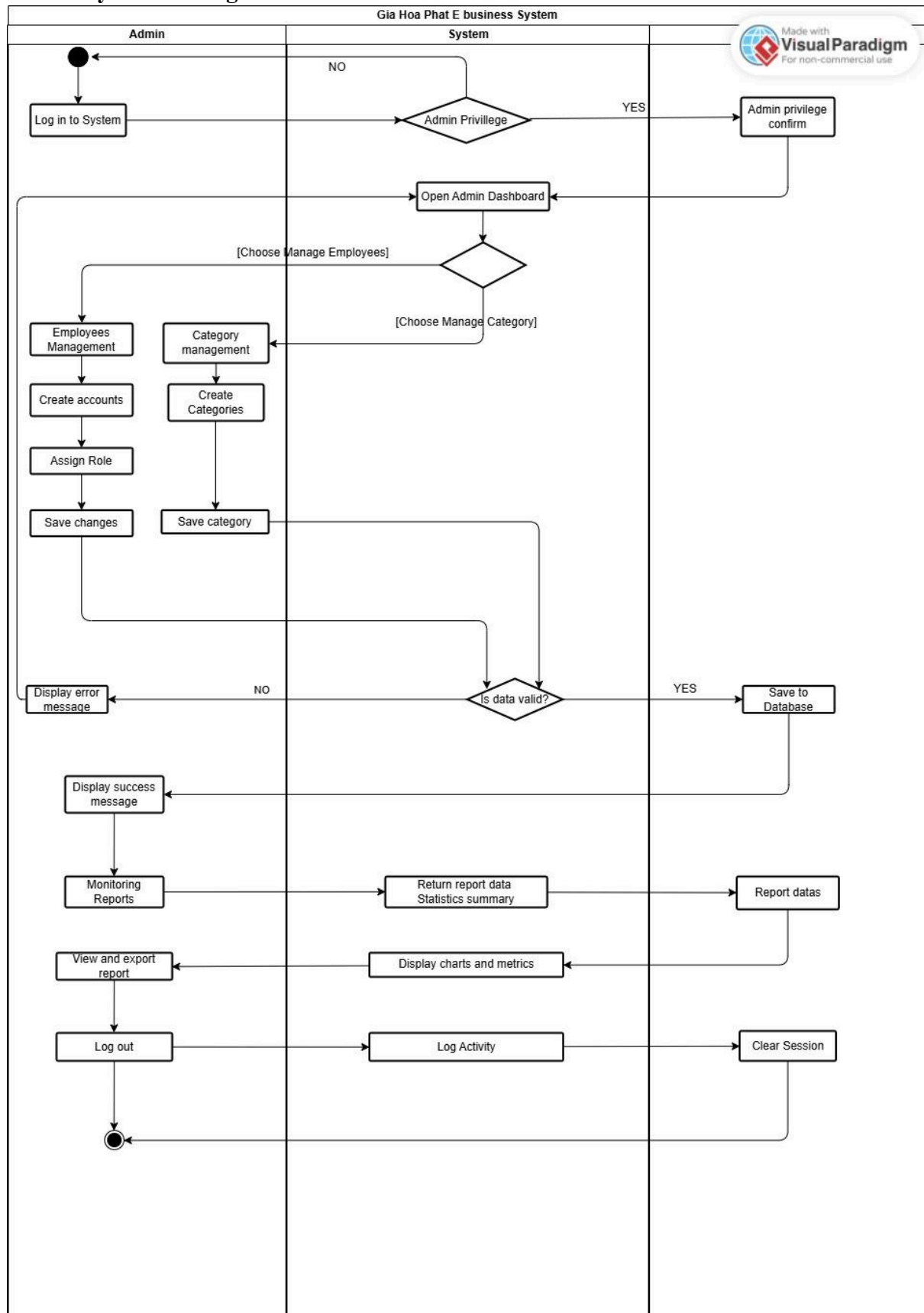
[This part describes the use cases, you can follow the table form as below]

Sample:

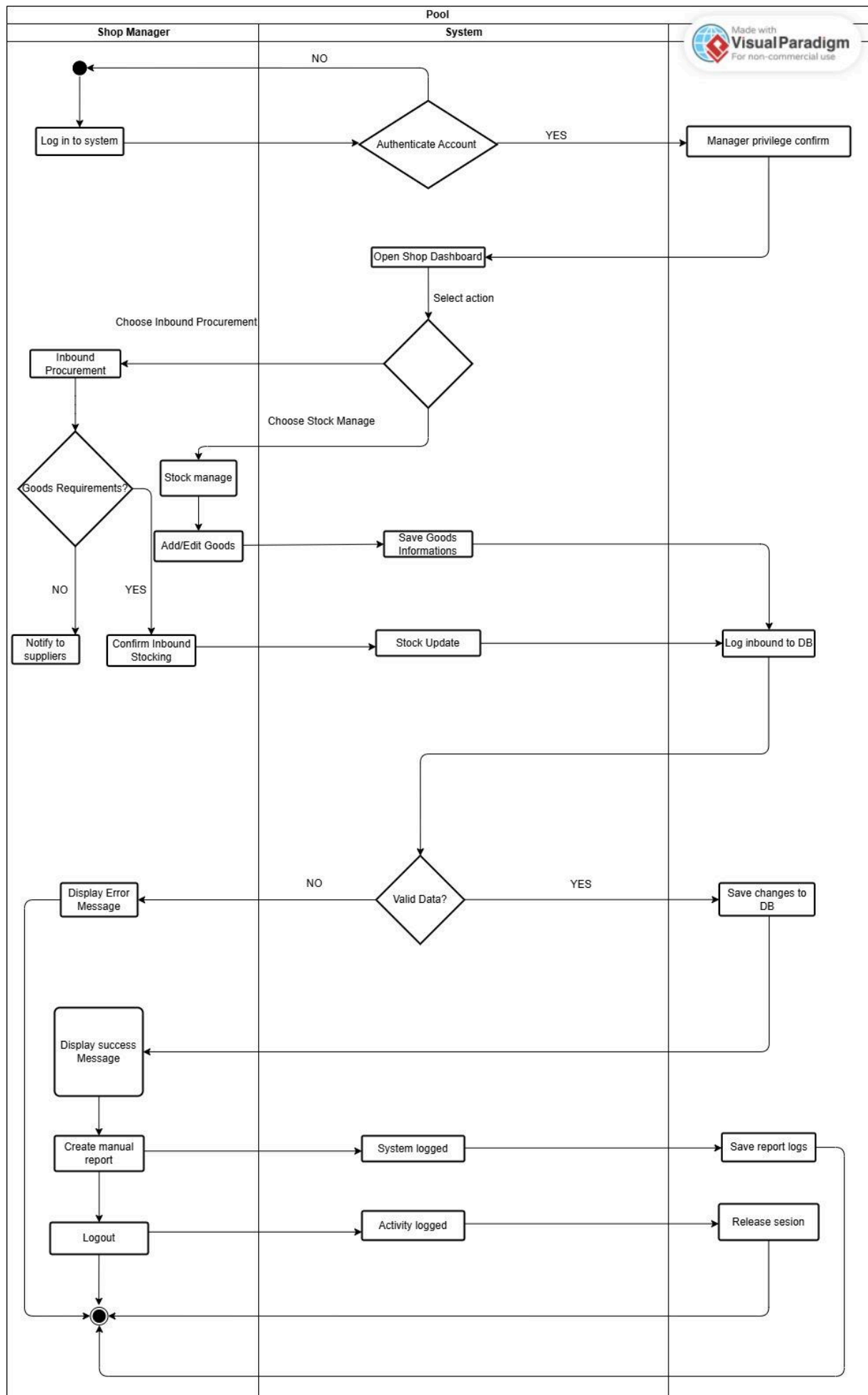
II.5.3.1. Staff Workflow



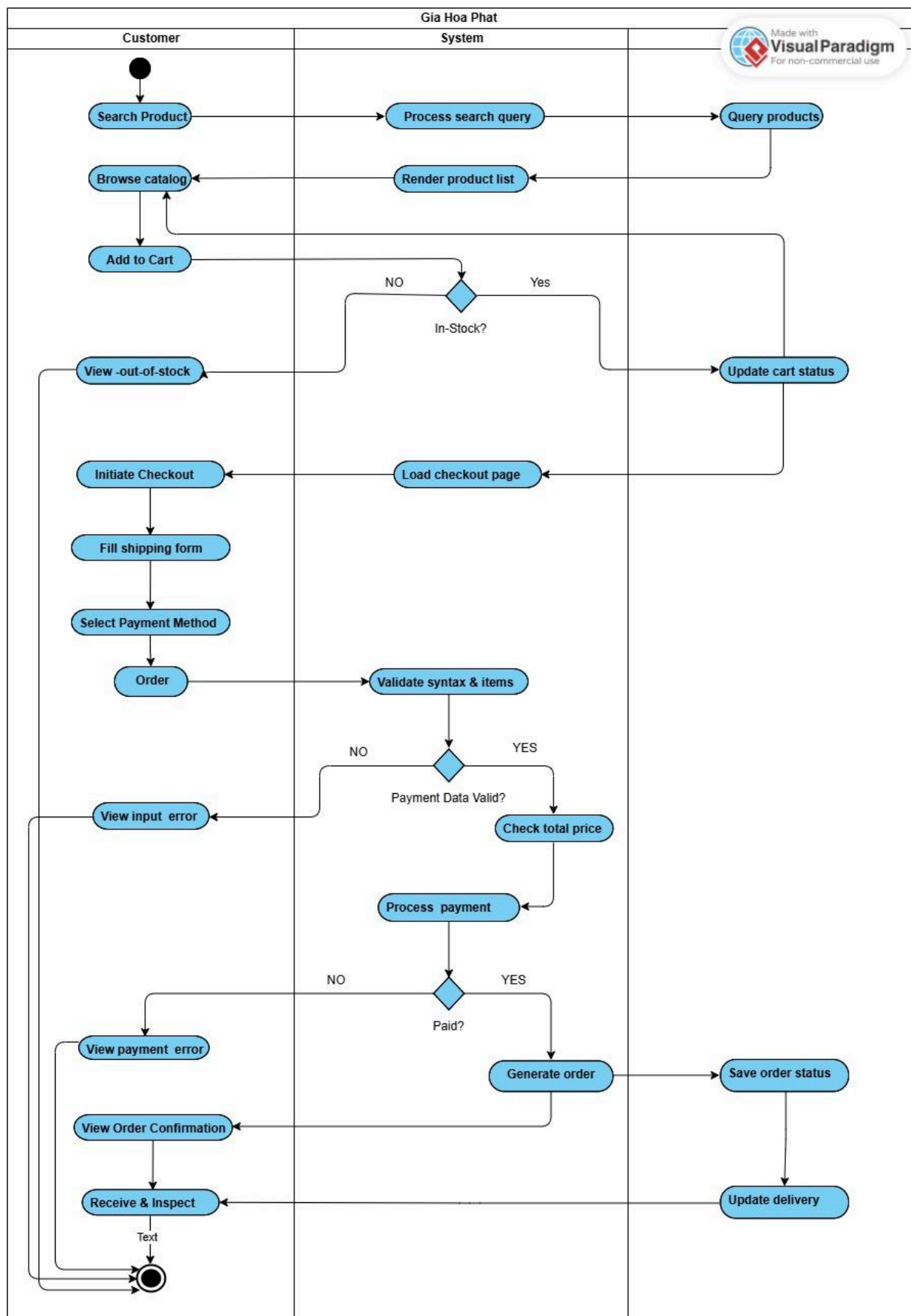
II.5.3.2.System Management



II.5.3.3.Shop Manager Workflow

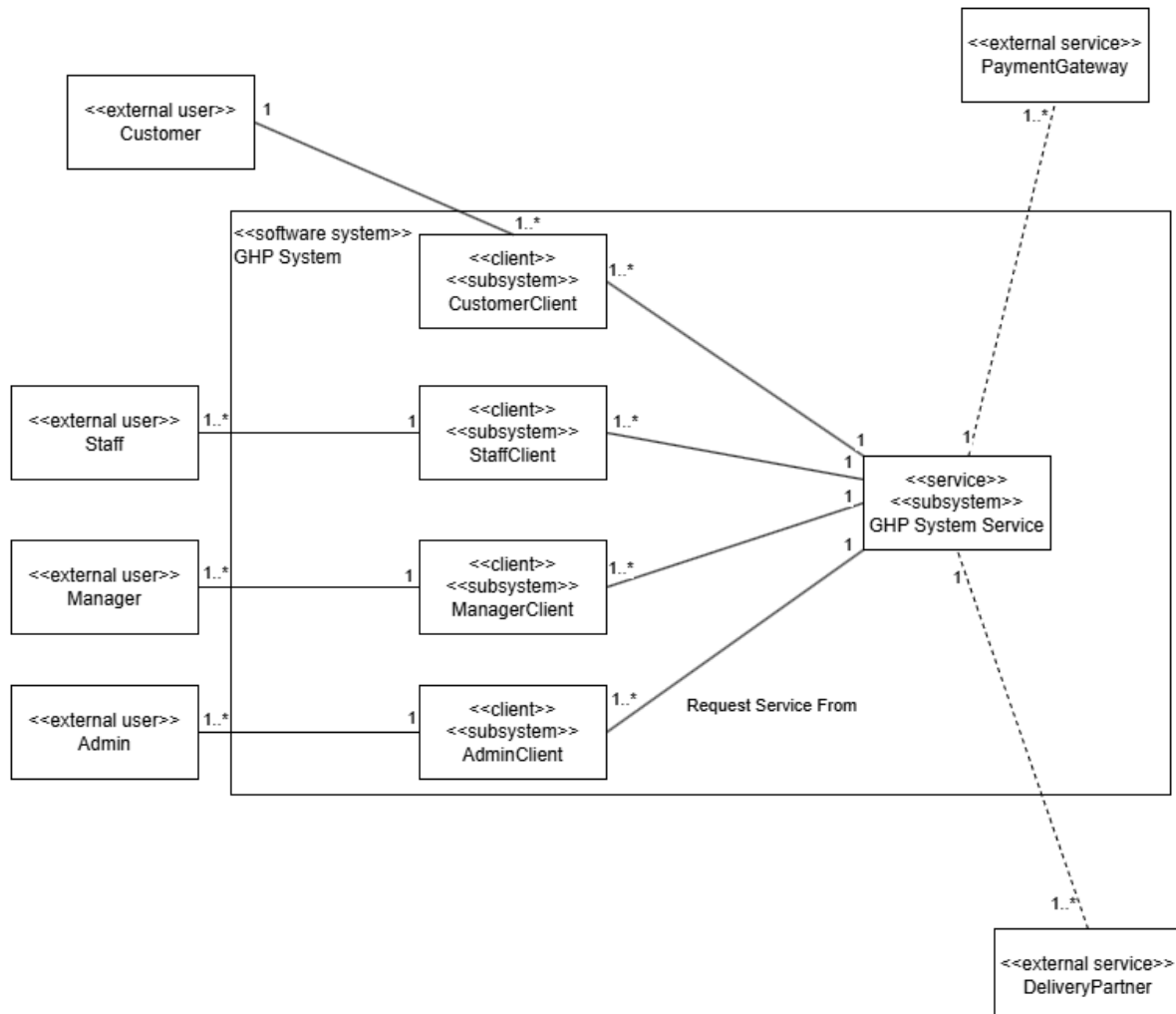


II.5.3.4.Buying Product Workflow(thêm tạo cart)

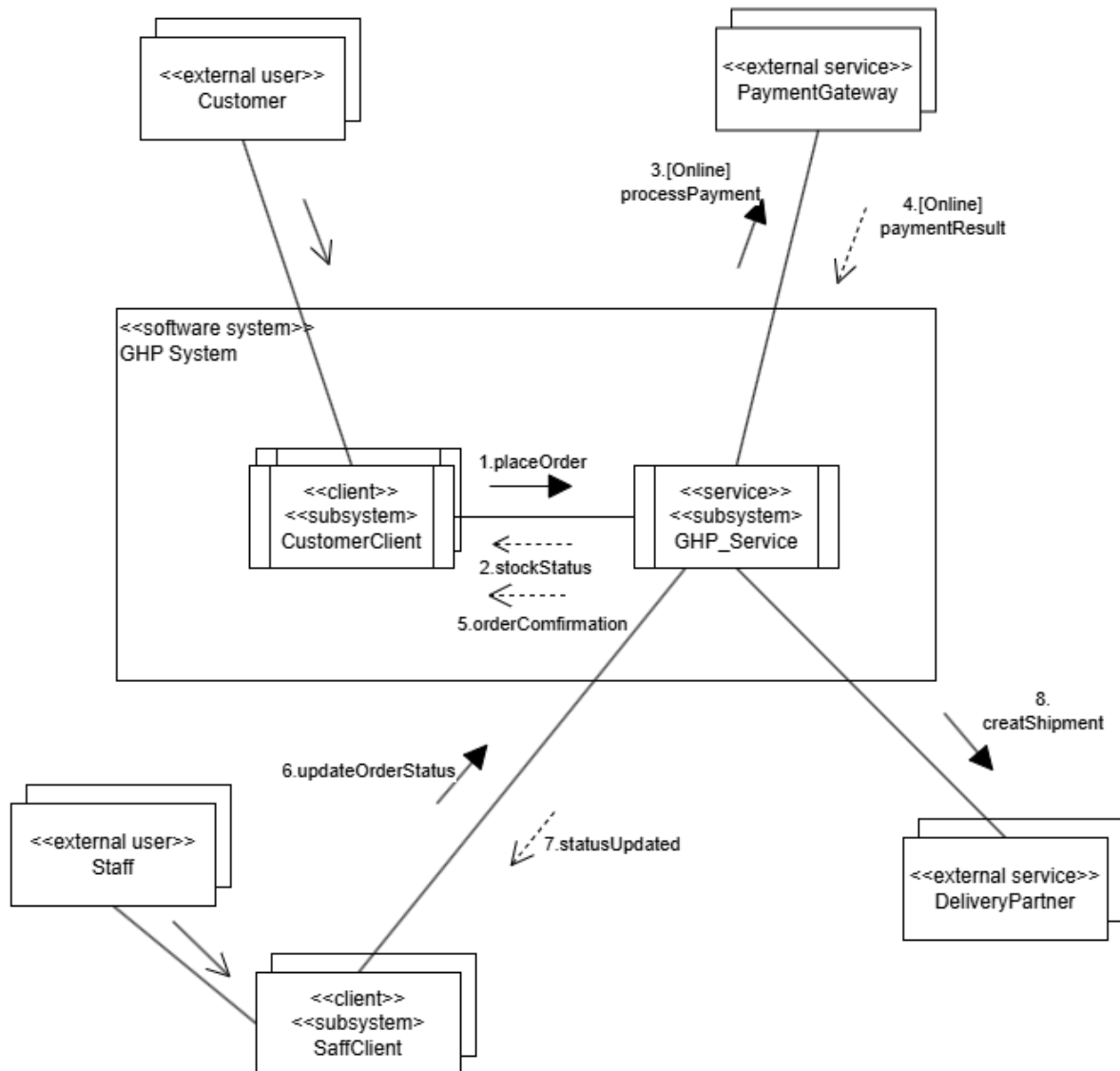


II.5.4 Architecture diagrams

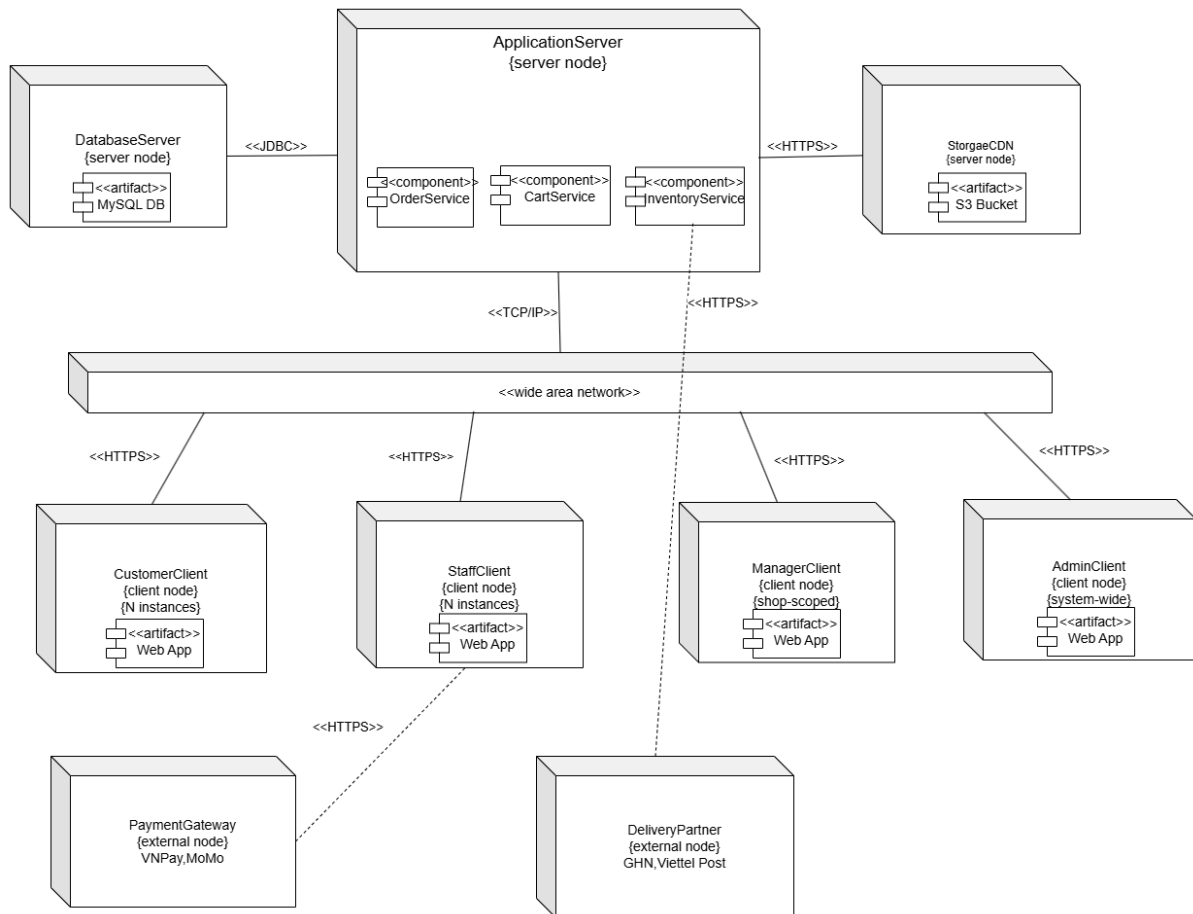
II.5.4.1 Structural view of client/server software architecture: high-level class diagram for GHP System



II.5.4.2 Dynamic view of client/server software architecture: high-level communication diagram for GHSystem



II.5.4.3 Deployment view of client/server software architecture: deployment diagram



II.6 Entity Relationship Diagram

[Provide the entity relationship diagram and the entity descriptions in the table format as below]

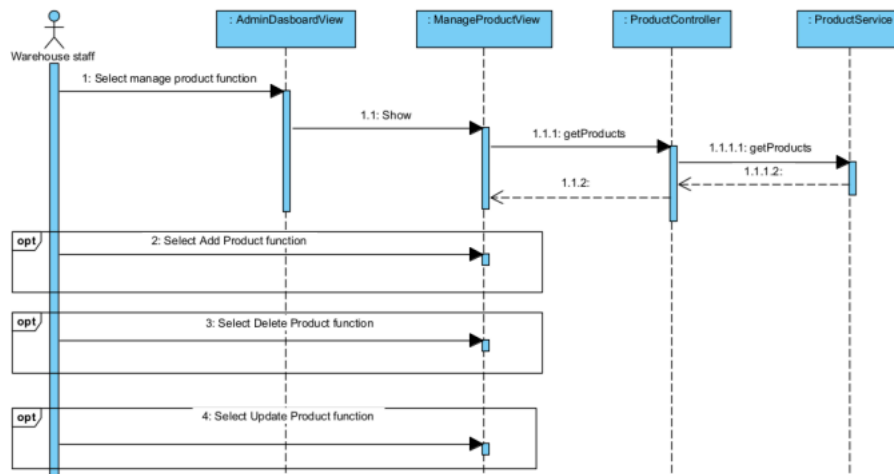


Figure II-1 Sequence Diagram for Manage Products Use Case

III.1.2. Communication Diagram

[Provide the Communication diagram(s) for the feature, see the sample below]

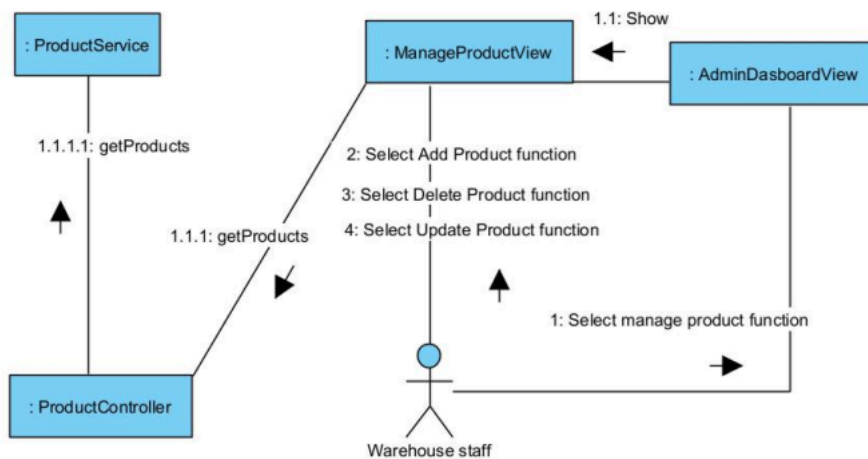


Figure II-7 Communication Diagram for Manage Products Use Case

III.2 State diagram

[Provide the State diagram(s) for the feature, see the sample below]

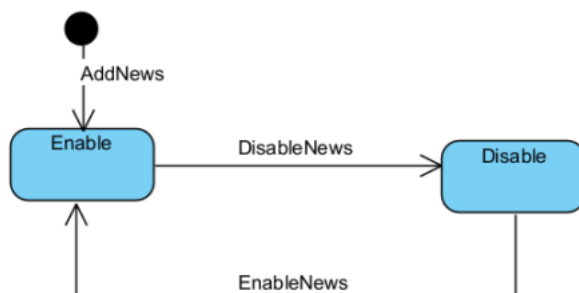


Figure II-13 State diagram model states of a News object

IV. Design specification

IV.1 Integrated Communication Diagrams

Sample

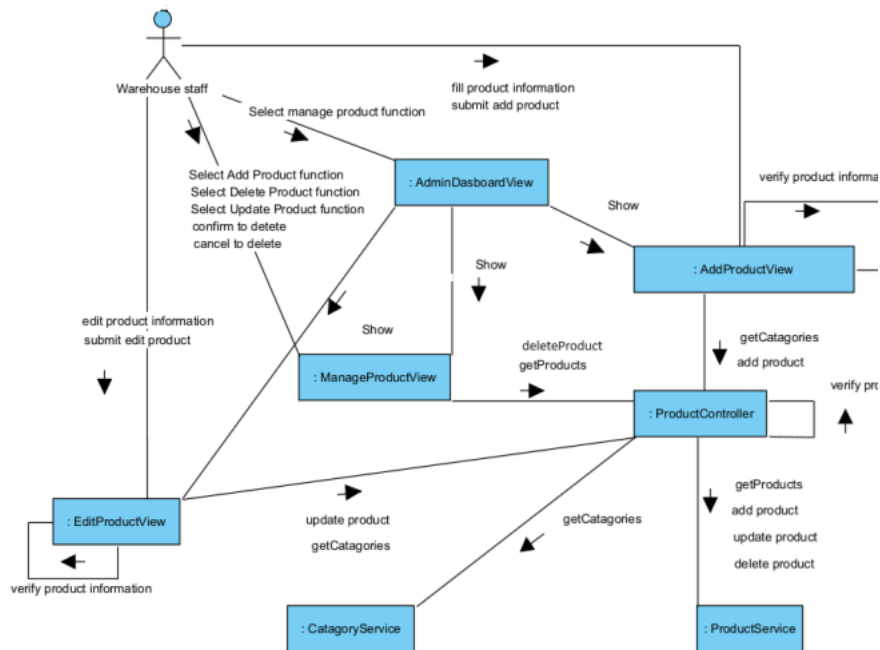


Figure III-1 Integrated communication diagram of Manage products, Add product, Edit product, Delete product.

IV.2 System High-Level Design

Sample

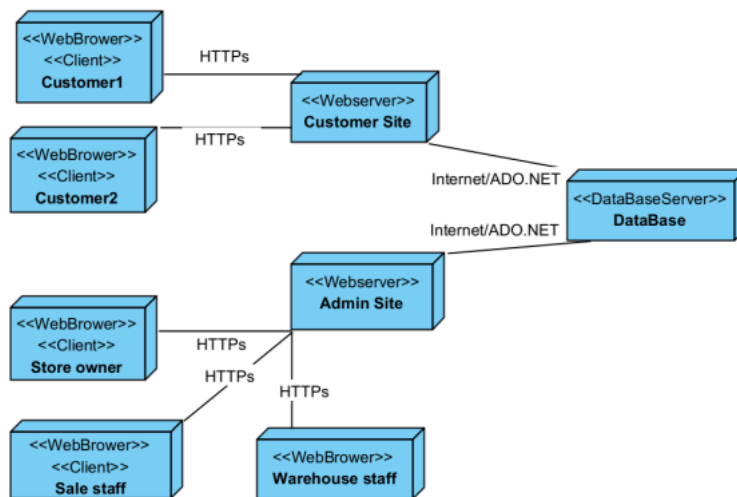


Figure III-2 Deployment diagram for the FSO system architecture

IV.3 Component and Package Diagram

IV.3.1 Component Diagram

Sample

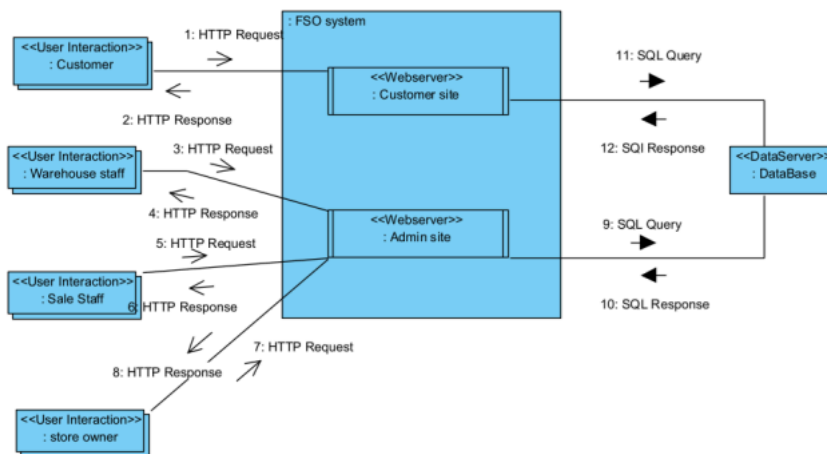
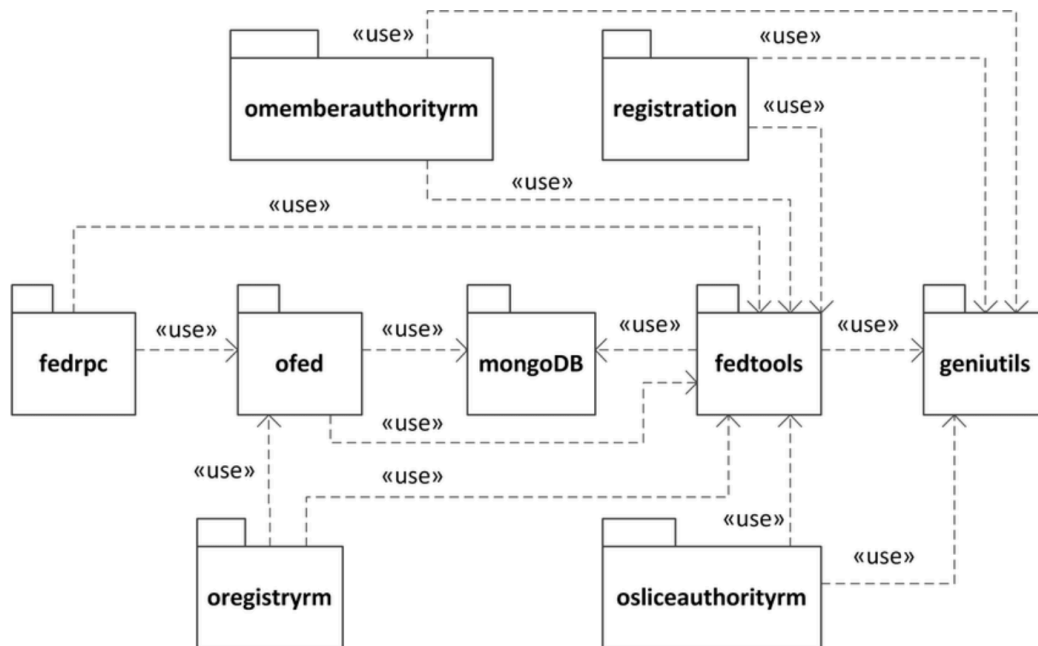


Figure III-4 Component diagram for the FSO system architecture

IV.3.1 . Package Diagram

[Provide the package diagram for each sub-system. The content of this section includes the overall package diagram, the explanation, package and class naming conventions in each package. Please see the sample & description table format below (please note: package names don't follow Java package naming convention yet)]

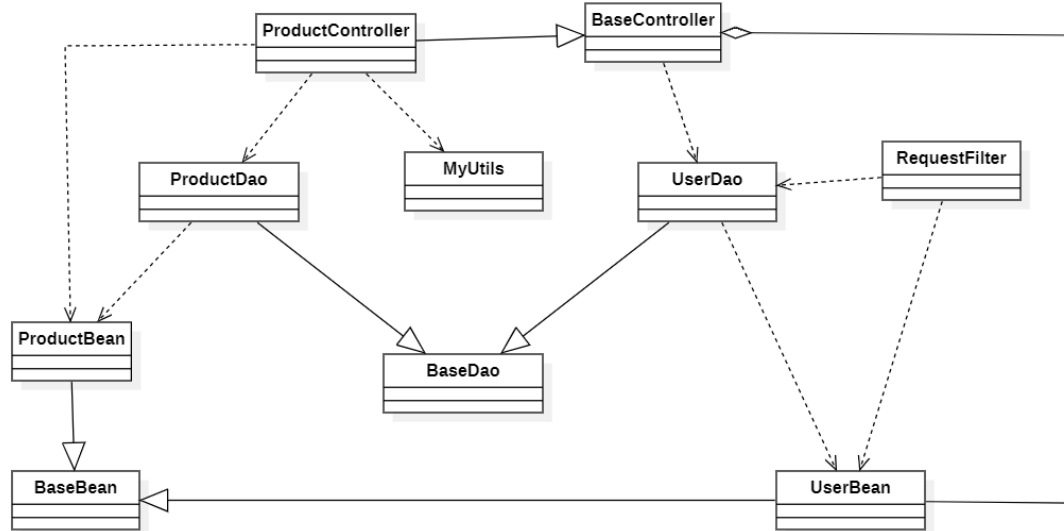


Package descriptions

No	Package	Description
01	omemberauthorityrm	<Description of the package>
02	registration	<Description of the package>
03	fedrpc	<Description of the package>
04	ofed	<Description of the package>
05	mongoDB	<Description of the package>
06	fedtools	<Description of the package>
07	...	

IV.4 Class diagram

[This part presents the class diagram for the relevant feature]



IV.5 Database Design

[Provide the tables relationship like example below – following MySQL database naming convention]

V. Implementation

V.1 Map architecture to the structure of the project

[Provide:

1. Overview of the Chosen Architecture

Briefly describe the architectural style (e.g., Layered, MVC, Service-Oriented Architecture)

*Explain **why it was selected** for this project (e.g., to meet non-functional requirements like maintainability, scalability, reusability)*

2. Mapping to Project Structure

*Present the **actual project folder/package structure***

Map each layer/component in the architecture to concrete project modules]

V.2 Map Class Diagram and Interaction Diagram to Code

[Demo the implementation of several features using design patterns]

VI. Applying Alternative Architecture Patterns

VI.1 Applying the (ex: **Service-Oriented Architecture (SOA)**) architecture

*[Redesign part of the system architecture using **Service-Oriented Architecture (SOA)** to improve **reusability, scalability, and modularity**.*

Students must provide:

1. Problem Identification

*Briefly describe which non-functional requirement(s) are **not fully supported** by the current architecture (e.g., NF-05: Reusability)*

2. SOA-Based Solution

*Reorganize system components into independent services
E.g., refactor FSOData into separate service modules*

3. Supporting Diagrams

Deployment Diagram

Updated Class Diagram

Component Diagram showing separate services]

VI.2 Applying Service Discovery Pattern in the service-oriented architecture

*[Support dynamic service registration and discovery in **SOA or microservice-based architecture** for **horizontal scalability** and ease of expansion.*

Students must provide:

1. Problem & Requirement

Explain the relevant non-functional requirement (e.g., NF-06: Scalability)

Describe the need for service discovery in a growing system (e.g., multiple branches, independent modules)

2. Service Discovery-Based Solution

*Integrate a **Service Discovery Pattern***

*Mention possible technologies: **Consul, Eureka, Kubernetes DNS***

3. Supporting Diagrams

Deployment Diagram (with multiple instances and registry)

Updated Class/Component Diagram including a Discovery Service]